Area and Perimeter Formulas

Area: length × width. Perimeter: all the way around.

A rectangle 8 m long and 5 m wide



Area: $8 \times 5 = 40$ sq m

Perimeter: $8 + 5 + 8 + 5 = 26$ m

or $2 \times 8 + 2 \times 5 = 26$: two long sides, two short.

Find a missing side from the area

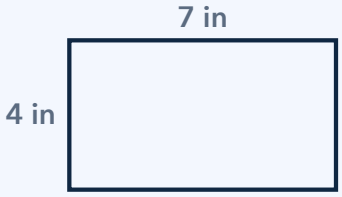
$8 \text{ cm} \times ? \text{ cm} = 48 \text{ sq cm}$

$48 \div 8 = 6$, so the width is 6 cm.

Find a missing side from the perimeter

Perimeter 30 ft, length 9 ft.
The two lengths are 18 ft, so the two widths are 30 minus 18 = 12 ft.
One width is half of 12: 6 ft.

Your turn!



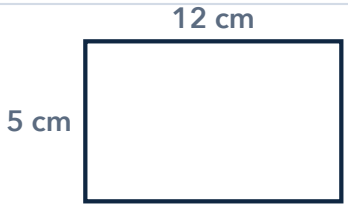
area: ____ sq in

perimeter: ____ in

Area and Perimeter Formulas: Practice

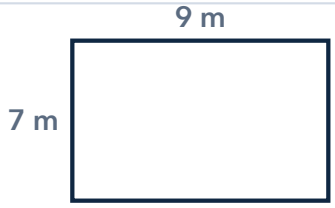
Area in square units, perimeter in units.

Part 1 Both formulas Two sides are labelled. The other two match them.

1) 

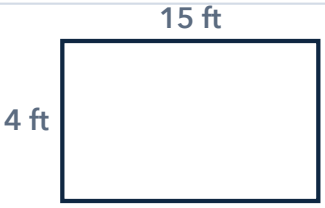
area: 12 × 5 = ☐ sq cm

perimeter: 2 × 12 + 2 × 5 = ☐ cm

2) 

area: 9 × 7 = ☐ sq m

perimeter: 2 × 9 + 2 × 7 = ☐ m

3) 

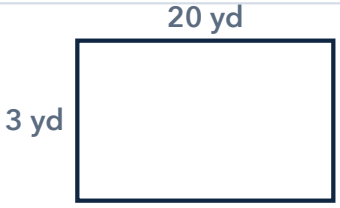
area: 15 × 4 = ☐ sq ft

perimeter: 2 × 15 + 2 × 4 = ☐ ft

4) 

area: 6 × 6 = ☐ sq in

perimeter: 2 × 6 + 2 × 6 = ☐ in

5) 

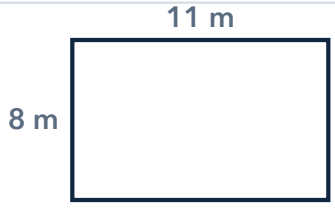
area: 20 × 3 = ☐ sq yd

perimeter: 2 × 20 + 2 × 3 = ☐ yd

Area and Perimeter Formulas: Practice (continued)

Part 1 Both formulas (continued)

6)



area: 11 × 8 = ☐ sq m

perimeter: 2 × 11 + 2 × 8 = ☐ m

Part 2 Area No help this time.

7) area: 14 × 6 = ☐ sq m

8) area: 23 × 4 = ☐ sq cm

9) area: 18 × 10 = ☐ sq ft

10) area: 25 × 3 = ☐ sq in

11) area: 16 × 7 = ☐ sq yd

12) area: 32 × 5 = ☐ sq m

13) area: 12 × 12 = ☐ sq cm

14) area: 40 × 6 = ☐ sq ft

15) area: 19 × 8 = ☐ sq in

16) area: 45 × 2 = ☐ sq m

Area and Perimeter Formulas: Practice (continued)

Part 3 Perimeter No help this time.

17) perimeter: $2 \times 14 + 2 \times 6 =$ ☐ m

18) perimeter: $2 \times 23 + 2 \times 4 =$ ☐ cm

19) perimeter: $2 \times 18 + 2 \times 10 =$ ☐ ft

20) perimeter: $2 \times 25 + 2 \times 3 =$ ☐ in

21) perimeter: $2 \times 16 + 2 \times 7 =$ ☐ yd

22) perimeter: $2 \times 32 + 2 \times 5 =$ ☐ m

23) perimeter: $2 \times 30 + 2 \times 20 =$ ☐ cm

24) perimeter: $2 \times 40 + 2 \times 6 =$ ☐ ft

25) perimeter: $2 \times 19 + 2 \times 8 =$ ☐ in

26) perimeter: $2 \times 45 + 2 \times 2 =$ ☐ m

Area and Perimeter Formulas: Practice (continued)

Part 4 Missing side from the area No help this time. Divide.

27) 9 cm × ☐ cm = 54 sq cm

28) 8 m × ☐ m = 72 sq m

29) 5 ft × ☐ ft = 65 sq ft

30) 7 in × ☐ in = 84 sq in

31) 12 yd × ☐ yd = 96 sq yd

32) 10 m × ☐ m = 120 sq m

33) 7 cm × ☐ cm = 91 sq cm

34) 15 ft × ☐ ft = 150 sq ft

Part 5 Missing side from the perimeter No help this time. Take away both lengths, then halve it.

35) P = 30 ft, length 9 ft: width = ☐ ft

36) P = 32 m, length 12 m: width = ☐ m

37) P = 50 cm, length 15 cm: width = ☐ cm

38) P = 50 in, length 20 in: width = ☐ in

Area and Perimeter Formulas: Practice (continued)

Part 5 Missing side from the perimeter (continued)

39) P = 30 yd, length 8 yd: width = ☐ yd

40) P = 80 m, length 25 m: width = ☐ m

Part 6 Squares No help this time. All four sides match.

41) square, side 9 in: A = ☐ sq in

42) square, side 11 m: A = ☐ sq m

43) square, side 15 cm: A = ☐ sq cm

44) square, side 9 in: P = ☐ in

45) square, side 11 m: P = ☐ m

46) square, side 15 cm: P = ☐ cm

Part 7 Area

47) area: 11 × 11 = ☐ sq in

48) area: 6 × 2 = ☐ sq cm

49) area: 13 × 12 = ☐ sq in

50) area: 23 × 5 = ☐ sq ft

Area and Perimeter Formulas: Practice (continued)

Part 7 Area (continued)

51) area: 16 × 8 = ☐ sq ft

52) area: 14 × 8 = ☐ sq in

53) area: 21 × 10 = ☐ sq ft

54) area: 13 × 10 = ☐ sq cm

55) area: 19 × 3 = ☐ sq in

56) area: 18 × 11 = ☐ sq in

57) area: 10 × 11 = ☐ sq ft

58) area: 12 × 9 = ☐ sq ft

59) area: 22 × 6 = ☐ sq cm

60) area: 25 × 9 = ☐ sq cm

61) area: 21 × 8 = ☐ sq ft

62) area: 15 × 4 = ☐ sq ft

Area and Perimeter Formulas: Practice (continued)

Part 7 Area (continued)

63) area: $18 \times 3 =$ ☐ sq in

64) area: $11 \times 8 =$ ☐ sq cm

Part 8 Perimeter

65) perimeter: $2 \times 24 + 2 \times 14 =$ ☐ ft

66) perimeter: $2 \times 21 + 2 \times 15 =$ ☐ ft

67) perimeter: $2 \times 25 + 2 \times 10 =$ ☐ cm

68) perimeter: $2 \times 6 + 2 \times 4 =$ ☐ in

69) perimeter: $2 \times 13 + 2 \times 15 =$ ☐ ft

70) perimeter: $2 \times 19 + 2 \times 13 =$ ☐ in

71) perimeter: $2 \times 6 + 2 \times 12 =$ ☐ ft

72) perimeter: $2 \times 26 + 2 \times 5 =$ ☐ in

73) perimeter: $2 \times 13 + 2 \times 12 =$ ☐ in

74) perimeter: $2 \times 10 + 2 \times 11 =$ ☐ in

Area and Perimeter Formulas: Practice (continued)

Part 8 Perimeter (continued)

75) perimeter: $2 \times 13 + 2 \times 6 =$ ☐ cm

76) perimeter: $2 \times 15 + 2 \times 9 =$ ☐ m

77) perimeter: $2 \times 17 + 2 \times 5 =$ ☐ cm

78) perimeter: $2 \times 25 + 2 \times 5 =$ ☐ cm

79) perimeter: $2 \times 17 + 2 \times 8 =$ ☐ ft

80) perimeter: $2 \times 26 + 2 \times 6 =$ ☐ in

81) perimeter: $2 \times 27 + 2 \times 9 =$ ☐ cm

82) perimeter: $2 \times 8 + 2 \times 15 =$ ☐ cm

Part 9 Remember? Problems from earlier packets.

83) $8 \text{ ft} =$ ☐ in

84) $7 \text{ kg} =$ ☐ g

85) 0.45 ☐ 0.5

86) $\frac{29}{100} =$ ☐

Area and Perimeter Formulas: Practice (continued)

Part 9 Remember? (continued)

87) $3 \times \frac{4}{5} = \frac{\square}{5}$

88) $64 \times 32 =$ ☐

Part 10 Word problems Draw the rectangle and label the sides.

89) A garden is 15 m long and 8 m wide. How much fence goes all the way around it?

Number sentence _____

Answer ☐ m

Draw or show your work

90) A rug is 11 ft long and 6 ft wide. What is its area?

Number sentence _____

Answer ☐ sq ft

Draw or show your work

91) A poster has an area of 72 sq in. It is 9 in wide. How long is it?

Number sentence _____

Answer ☐ in

Draw or show your work

Area and Perimeter Formulas: Practice (continued)

Part 10 Word problems (continued)

92) A square room has sides of 12 ft. How much carpet covers the floor?

Number sentence _____

Answer ____ sq ft

Draw or show your work

93) Challenge: A pen has 40 m of fence around it. It is 12 m long. How wide is it?

Number sentence _____

Answer ____ m

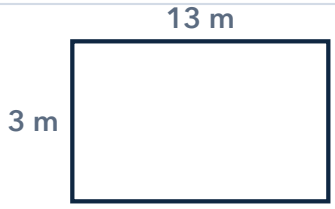
Draw or show your work

Check-In: Area and Perimeter Formulas

Show what you know. No help this time. Score: _____ / 9

Part 1 Both formulas

1)



area: 13 × 3 = ☐ sq m

perimeter: 2 × 13 + 2 × 3 = ☐ m

Part 2 Area

2) area: 35 × 4 = ☐ sq cm

3) area: 47 × 4 = ☐ sq m

Part 3 Perimeter

4) perimeter: 2 × 17 + 2 × 5 = ☐ in

5) perimeter: 2 × 20 + 2 × 8 = ☐ ft

Part 4 Missing side

6) 7 m × ☐ m = 70 sq m

7) 12 m × ☐ m = 84 sq m

Part 5 Missing side from the perimeter

8) P = 68 m, length 26 m: width = ☐ m

9) P = 46 m, length 14 m: width = ☐ m